Ge'ez Morphological Analyser (ByT5): Training Report, Run A

geez-morphology-model commit a1ad2b6 · geez-dataset commit 04f45a7 · Kaggle notebook esubechekol/geez-byt5-gpu-v2-a · report built 18 October 2026

Result. Run A trained to completion (10,980 steps, 10 epochs). The selected checkpoint analyses held-out Ge'ez words with **92.1% root accuracy**, **69.6% exact full analysis** (root and every feature) and **86.3% part-of-speech accuracy** on 1,736 held-out validation words whose roots never occur in training. This model is the planned label source for the next translation experiments. These are validation numbers used for model selection; the separate held-out test set (2,310 pairs) has not been scored yet.

1. What the model does

The analyser reads one Ge'ez word, character by character (bytes), and writes its morphological analysis as text. It is trained on three tasks at once, marked by a task prefix: **analyse** (word → root + pattern features), **generate** (root + features → word) and **part of speech** (word → word class). For example, ሰገዱ should be analysed as root s-g-d with perfective, 3rd person, plural, masculine features, and tagged as a verb.

Item	Value
Architecture	ByT5-small encoder–decoder (T5, gated-GELU): 12 encoder / 4 decoder layers, d_model 1472, feed-forward 3584, 6 heads, byte vocabulary 384
Run type	A = fine-tuning from the public ByT5-small checkpoint (no Ge'ez pretraining). Run B adds Ge'ez span-corruption pretraining first.
Hardware	Kaggle, 2 × NVIDIA T4, PyTorch 2.11.0+cu128 DDP, fp32
Batch	32 rows × 2 GPUs × 8 accumulation = 512 rows per step; 1,098 steps per epoch
Schedule	10,980 steps = 10 epochs; warm-up 500 steps; logged learning rate peaks at 1.0e-03 and decays to 1e-04
Model selection	every 1,000 steps on the validation set; score = mean of exact-analysis accuracy and root accuracy; best checkpoint kept in best/
Wall-clock	three Kaggle sessions of about 11 h each (session 1: 11.3 h, steps 1–3,676; session 2: steps 3,677–7,530; session 3: 10.8 h, steps 7,531–10,980 plus the final report), about 33 h in total, at a median 0.092 steps/s

2. Training data

All data come from the author's geez-dataset (commit 04f45a7). Before training, the script checks that the data load exactly as expected and that no validation or test root, and no validation or test word form, appears in training.

Source	Word pairs	What it is
synthetic	118,131	forms generated from checked roots and paradigm rules
unified_book	10,002	forms transcribed from printed grammars and books (unified table)
gold_nominal	6,168	hand-checked noun/adjective paradigm cells
gold_verb	1,727	hand-checked verb paradigm cells
dictionary	1,080	dictionary headwords with checked roots
extra_synthetic	247	additional checked generated cells
dictionary_verb	59	dictionary verbs with checked roots
word_without_root	32	function words labelled as rootless
total	137,446	expanded to 562,403 training samples over the three tasks

Leakage audit (from the run). Training: 562,403 samples, 1,946 distinct roots. Validation: 1,736 pairs, 147 roots; test: 2,310 pairs, 217 roots. Training pairs sharing a root, a word form, or a spelling-variant root with validation or test: **0 in every check** (RESULT: CLEAN).

Class balance. Verbs dominate the training samples: verb 422,691 (75.2%), adjective 68,400 (12.2%), noun 56,196 (10.0%), nominal 9,200 (1.6%); all other classes together 1.1%. Synthetic forms are 86% of word pairs.

3. Training curve

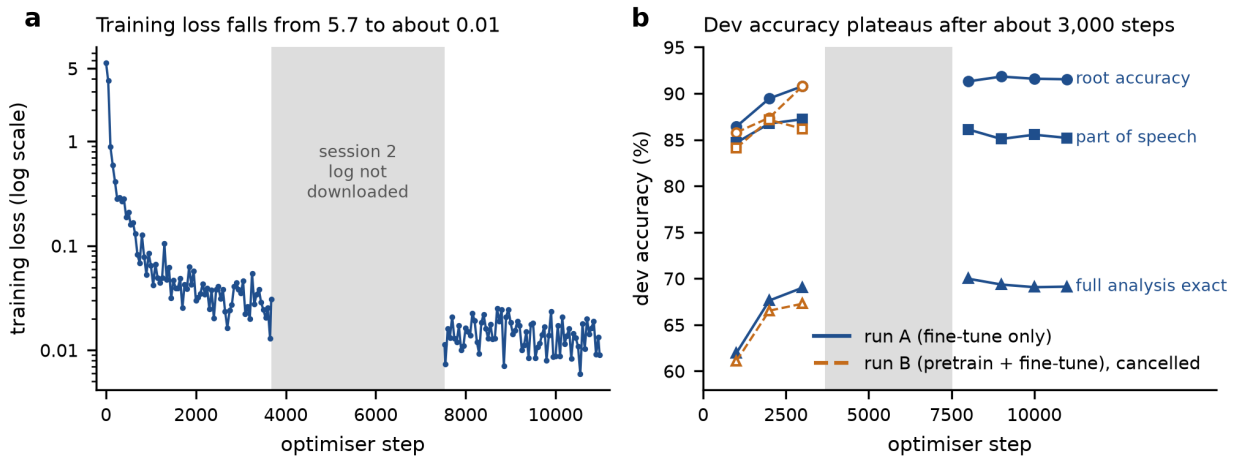


Figure 1. (a) Training loss per logged step. (b) Validation accuracy every 1,000 steps for run A (solid) and the cancelled run B (dashed, steps 1,000–3,000 only). The grey band marks session 2 (steps 3,677–7,530), whose log and evaluations have not been downloaded; the best checkpoint was selected inside this band (selection score 0.8088, above every logged evaluation).

Step	1,000	2,000	3,000	8,000	9,000	10,000	10,980
Root accuracy	86.4%	89.5%	90.8%	91.3%	91.8%	91.6%	91.5%
Exact analysis	61.9%	67.6%	69.0%	70.0%	69.4%	69.1%	69.1%
Part of speech	84.7%	86.8%	87.2%	86.1%	85.1%	85.5%	85.2%
Selection score	0.7416	0.7854	0.7990	0.8065	0.8059	0.8033	0.8033

Validation accuracy rises quickly to step 3,000 and then changes by less than one point. The final step (10,980) scores 0.8033, slightly below the best checkpoint (0.8088), so the last epochs mainly fit the training set further (loss about 0.01) without improving held-out words. Run B matched run A at step 3,000 on root accuracy (90.8% both) but was lower on exact analysis (67.3% vs 69.0%); B was cancelled, so no conclusion about pretraining is drawn.

4. Results of the selected checkpoint

Validation words come from two sources: **gold paradigm** cells (hand-checked tables) and **book/unified** rows (forms transcribed from printed sources). Root accuracy counts the predicted root only; exact analysis requires the root and every feature to be right.

Source / class	Words	Root	Exact analysis	Part of speech
gold paradigms				
adjective	503	92.6%	74.0%	85.3%
noun	440	88.6%	76.1%	84.1%
verb	221	95.5%	77.8%	99.1%
book / unified rows				
verb	270	91.9%	76.3%	97.4%
adjective	193	95.3%	46.1%	90.2%
noun	61	98.4%	37.7%	45.9%
infinitive	20	95.0%	15.0%	5.0%
particle	9	88.9%	66.7%	88.9%
pronoun	7	57.1%	0.0%	28.6%
demonstrative	4	75.0%	0.0%	25.0%
preposition	4	100.0%	25.0%	25.0%
numeral	3	33.3%	33.3%	66.7%
nominal	1	100.0%	100.0%	100.0%
all validation words	1,736	**92.1%**	**69.6%**	**86.3%**

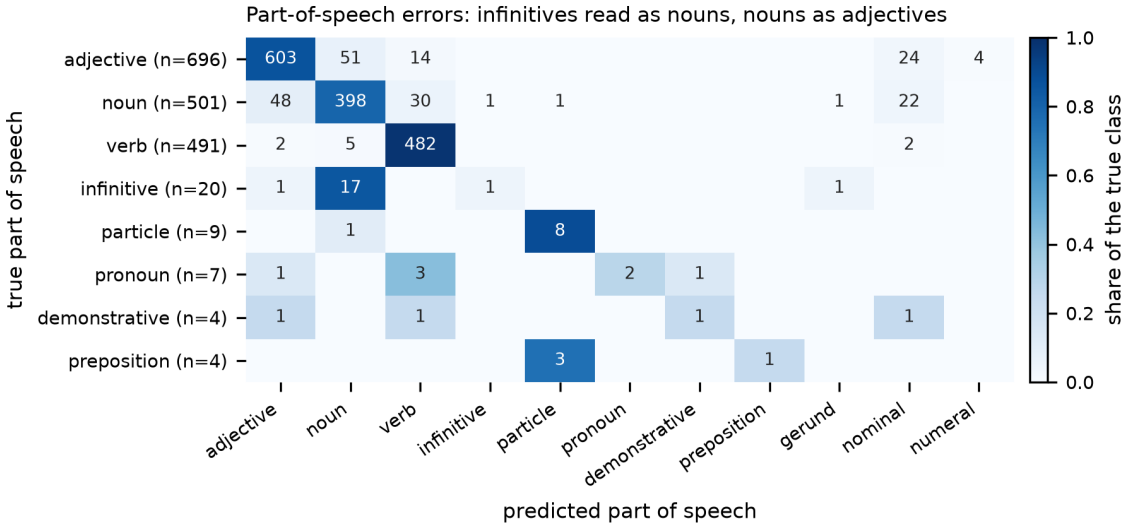


Figure 2. Part-of-speech confusion on validation words (classes with at least 4 words; counts in cells, colour = share of the true class).

5. What the numbers mean

• **Roots are reliable.** 92% of held-out words get the right root, including words whose root was never seen in training. This is the property the translation memory needs, because memory slots are tied to roots.
• **Full analyses are harder.** 70% are exactly right. Gold paradigm cells are easier (74–78% by class) than book rows (15–76% for classes with at least 20 words), which are more varied and closer to running text.
• **Weak spots.** Infinitives are almost always tagged as nouns (17 of 20); nouns are tagged as adjectives in 48 of 501 cases; pronouns, prepositions and demonstratives are rare in training (under 0.2% of samples each) and are often wrong.
• **Over-fitting is mild.** Training loss keeps falling to about 0.01 while validation stays flat after about 3,000 steps;

about a third of the compute would have given almost the same model.

6. Limitations

1. All numbers are validation numbers, and the same set chose the checkpoint, so they are slightly optimistic. The 2,310-pair test set has not been scored.
2. The session-2 log (steps 3,677–7,530), including the evaluation that selected the best checkpoint, has not been downloaded.
3. The model weights are not in the local folder: out_v6/best/ holds only the configuration and tokenizer files.
4. Validation words are dictionary and paradigm forms, not words in running sentences; accuracy on AGE sentences can be lower, especially for names and clitics.
5. Run B (with Ge'ez pretraining) was cancelled, so the effect of pretraining is unknown.

7. Next steps

1. Download best/ with its weights and the session-2 log (eval_log.jsonl).
2. Score the held-out test set once with the selected checkpoint.
3. Re-label the AGE corpus with this model and compare the new labels with the older analyser labels used so far in the translation experiments (coverage, agreement, root changes), then rebuild the memory label caches from it.
4. Use its part-of-speech output as an additional signal for translation (for example, a part-of-speech slot bank or word-order features).
5. Run B on TPU once the TPU runtime is stable, and compare A and B at matched steps.